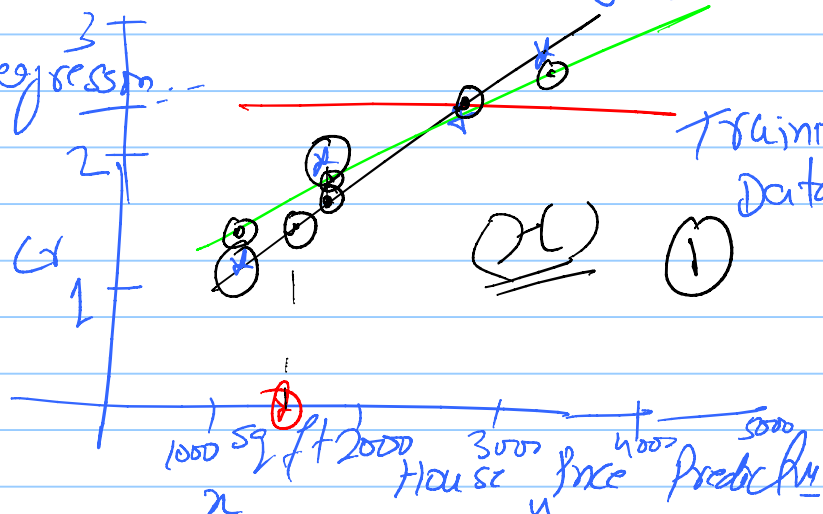


Machine Learning

Aug 5, 2025

Linear Regression



Area of House	Price of House
1. 1257	1.1 cr
2. 1783	1.9 cr
3. 2489	2.2 cr

$$x^n \rightarrow \boxed{M} \rightarrow \hat{y} \approx y$$

$$\equiv \{(x^{(i)}, y^{(i)})\}_{i=1}^m$$

$$x, y$$

$$x^{(i)}, x \in \mathcal{X} \equiv \mathcal{R}^n$$

$$y \in \mathcal{Y} \subseteq \mathcal{R}$$

$$h_\theta: x \rightarrow \hat{y}$$

$$x \rightarrow y?$$

$$h_\theta \in \mathcal{H}$$

4 | 3187

3.10.17

$\frac{h_0 \in \mathbb{R}}{h_0} \in \mathbb{R}$

$$h_0(x^{(u)}) = \hat{y}^{(u)}$$

$$h_0 \equiv (\theta_0, \theta_1, \dots, \theta_n) \text{ line}$$

$n \in \mathbb{R}^n$

$$\hookrightarrow h_0 \equiv (\theta_0, \theta_1, \dots, \theta_n) \text{ hyperplane}$$

(Prediction)

$$\textcircled{2} J(\theta) = \frac{1}{2m} \sum_{i=1}^m [y^{(i)} - h_0(x^{(i)})]^2$$

$\hookrightarrow$ (M.S.E) cross metrics

Mean Squared Error

$$\textcircled{3} \arg \min_{\theta} J(\theta)$$

(Optimization Algorithm)